

WASP-50b

R-0740

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_231219 · created 2026-07-28T01:45:30+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460297.72 +/- 0.098 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.05
Transit depth (Rp/Rs)^2	3.02 +/- 1.75 [%]
Orbital Inclination (inc)	88.21 +/- 2.84
Airmass coefficient 1 (a1)	4.33 +/- 3.22
Airmass coefficient 2 (a2)	-0.63 +/- 0.28
Scatter in the residuals of the lightcurve fit is	73.06 %
Best Comparison Star	#1 - [349, 401]
Optimal Aperture	1.64
Optimal Annulus	4.81
Transit Duration (day)	0.045 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.05 vs 0.1404 (deviation 0.51σ)
Duration fit vs published	0.045 d vs 0.0754 d (deviation 0.63σ)
Mid-time O–C	-54.5 min
Depth SNR	2.7
Residual scatter	73.06 %

Light curve

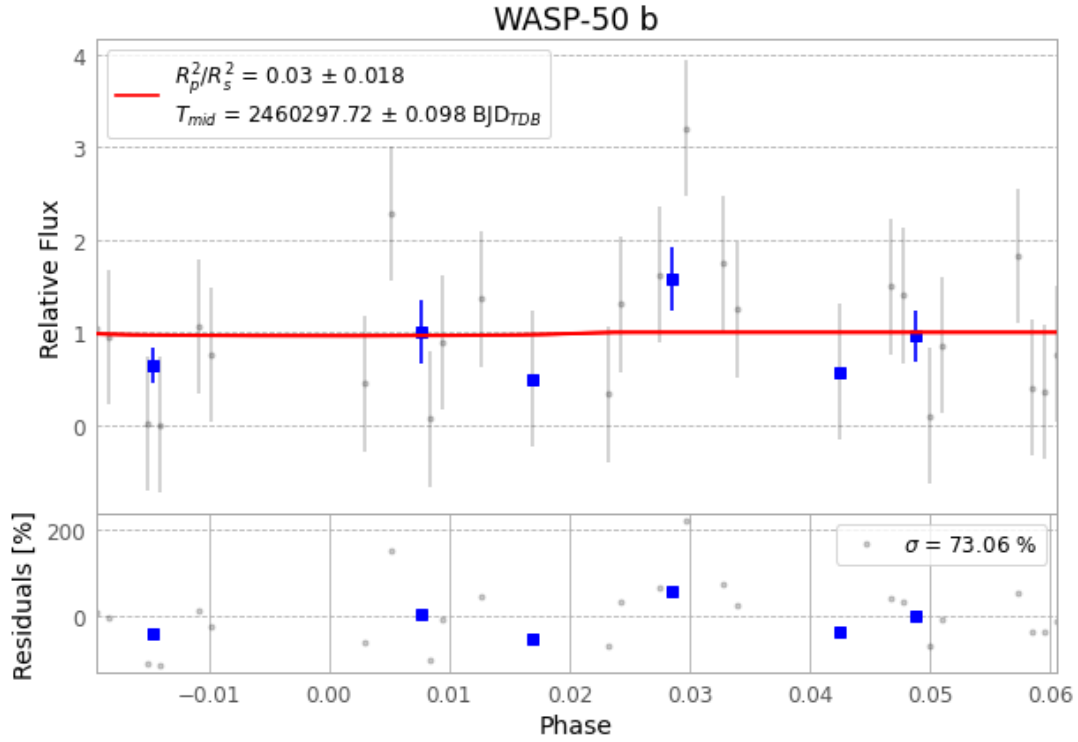


Figure 1 — FinalLightCurve_WASP-50 b_2023-12-18.png (SHA-256 5a8b9ed38ce0a23f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [298, 389], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 96a8adf621f4641f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), WASP-50231219041215.FITS → WASP-50231219080015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-231219.log (b8bf5ffcc451...), FinalLightCurve_WASP-50 b_2023-12-18.png (5a8b9ed38ce0...), FinalLightCurve_WASP-50 b_2023-12-18.csv (c43331390b82...)

Compute resources

Wall time 180.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 01:45Z.